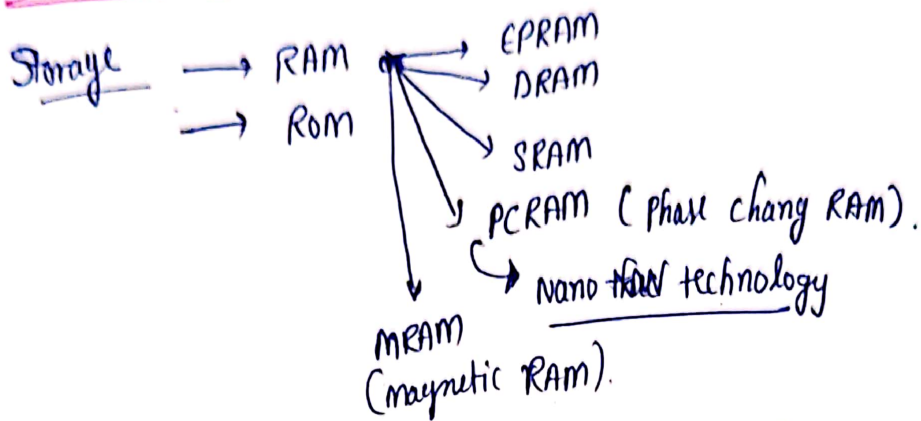
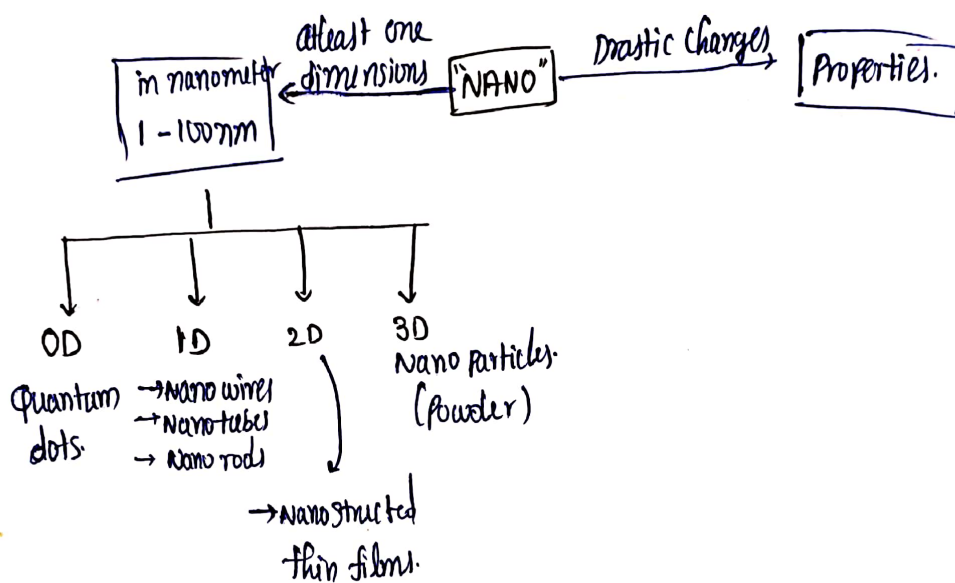


Nishaad D. Pandey NANO DEVICE ENGINEERING

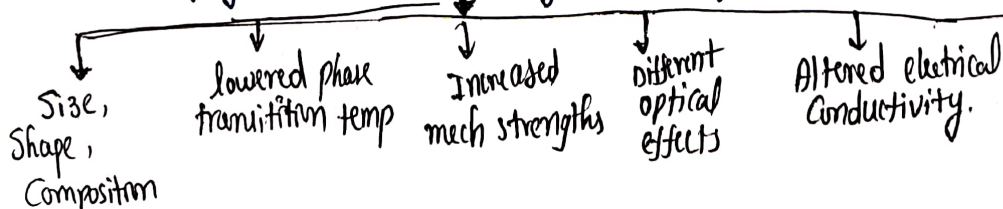


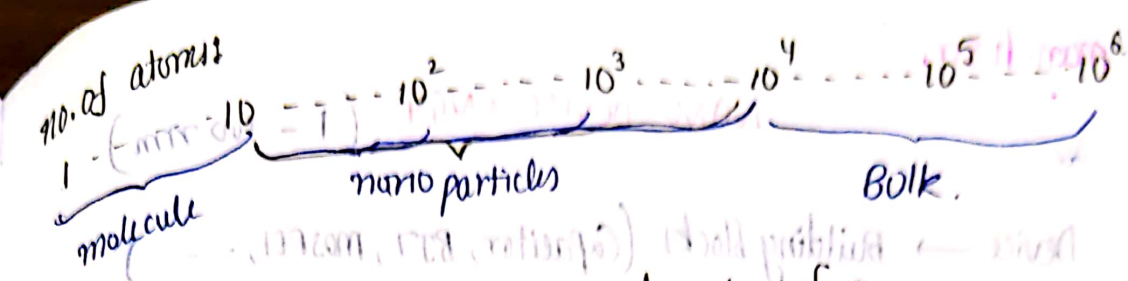
Data Storage 'vs' Information Storage devices. ?



Drastic change in the properties:

1. Increased surface to volume ratio.
2. High concentration of atoms/molecules at the surface
3. Dominant surface forces.
4. Plasmonic resonance effects.
5. Confined energy states [quantum confinement].
6. Drastic physical chemical biological changes.





Q: A spherical particle $\rightarrow$ 100 nm diameter then
 (i) Volume ($\frac{4}{3}\pi r^3$) (ii) surface area ($4\pi r^2$)

$$V = \frac{4}{3}\pi (50 \times 10^{-9})^3$$

$$V = 5.22 \times 10^{-22} \text{ m}^3$$

$$S = 4\pi (50 \times 10^{-9})^2$$

$$S = 3.14 \times 10^{-14} \text{ m}^2$$

$$\frac{S}{V} = \frac{3.14 \times 10^{-14}}{5.22 \times 10^{-22}} \text{ m}^2/\text{m}^3 \Rightarrow \frac{S}{V} = 6 \times 10^7 \text{ m}^{-1}$$

"SURFACE EFFECTIVE"

If diameter $\rightarrow$ 100 m.

$$V = 5.22 \times 10^{-22} \times \frac{1}{10^{-27}}$$

$$V = 5.22 \times 10^5 \text{ m}^3$$

$$S = 3.14 \times 10^{-14} \times \frac{1}{10^{-18}}$$

$$S = 3.14 \times 10^4 \text{ m}^2$$

(NANO)

(BULK)

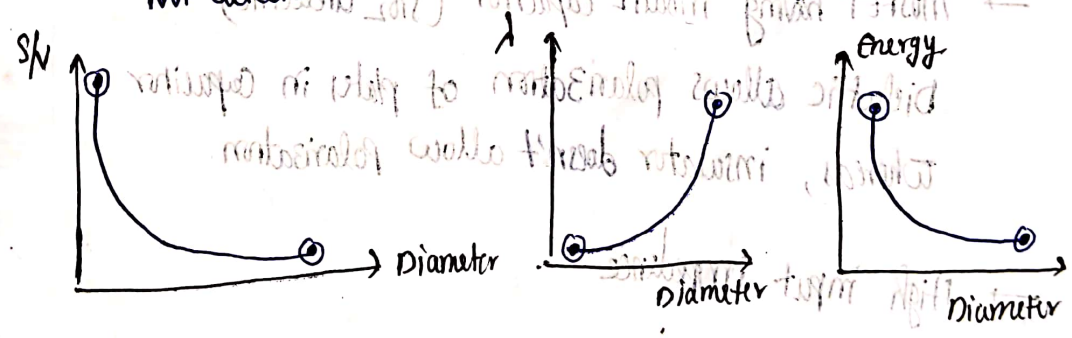
$$\frac{S}{V} = 0.06 \text{ units}$$

$$E_1 = \frac{hc}{\lambda} = \frac{hc}{100 \times 10^{-9}} = \frac{hc \times 10^7}{1}$$

Not detectable

$$E_2 = \frac{hc}{\lambda} = \frac{hc}{100} = \frac{hc \times 10^{-2}}{1}$$

Radio regm.



Changed Magnetic Properties.
 (Ferro, Dia, Anti, Ferri, Paramagnetic (Super para)?)

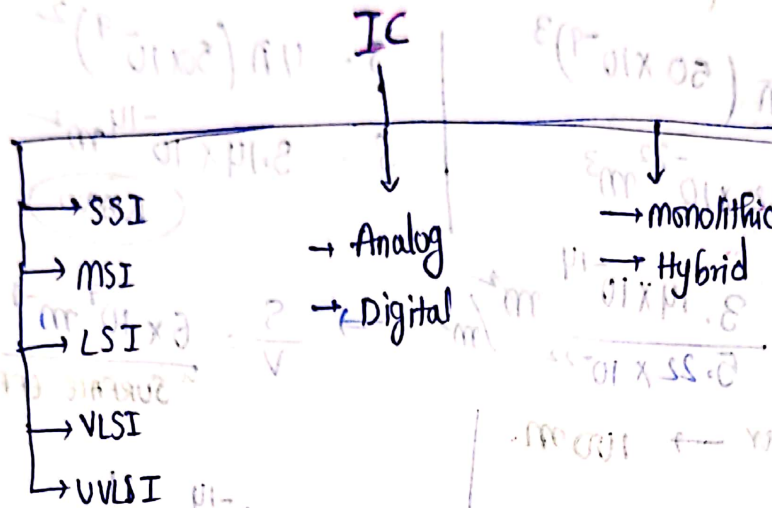
Effective Surface Chemical reactions.

manish srrs

NAND DEVICE ENGG (1 - 100 nm)

Device $\rightarrow$ Building blocks (Capacitor, BJT, MOSFET, ...)

Circuit $\rightarrow$ Connecting devices



IC fabrication:

Silicon wafer $\rightarrow$ on which fabrication takes place

$\rightarrow$ MOSFET $\gg$ BJT

$\rightarrow$ mosfet takes less place

$\rightarrow$ ~~mosfet~~ MOSFET has less power consumption than BJT

$\rightarrow$ MOSFET is voltage controlled device

$\rightarrow$ MOSFET having inbuilt capacitor (SiO_2 dielectric)

Dielectric allows polarization of plates in capacitor
Whereas, insulator doesn't allow polarization.

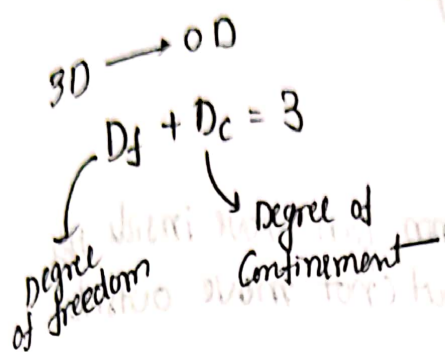
$\rightarrow$ High input impedance

$\rightarrow$ Easy fabrication.

Drawback: "Low speed."

Energy levels and quantum confinement effects

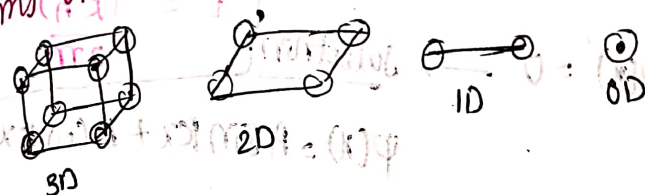
NISHAD SIR:-



	D_f	D_c
3D	3	0
2D	2	1
1D	1	2
0D	0	3

Q: If 0D nano material is having 0D then where is the particle going or moving? [Homework]!

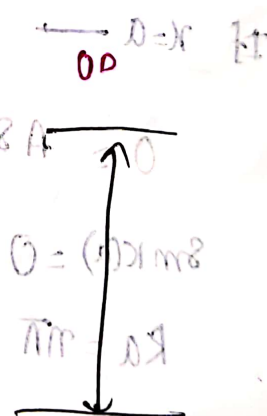
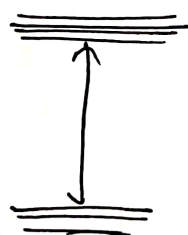
In case of Confinement we are decreasing the no. of atoms (on the surface).



Energy spectrum of 3D: It is having more no. of electrons and more no. of energy levels and less Band gap.

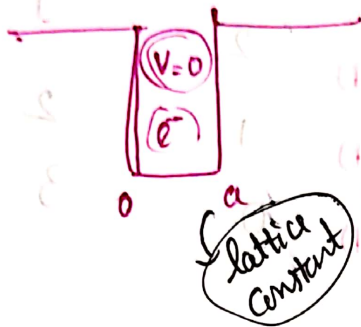


Energy spectrum of 2D:



As the no. of atoms decreases the no. of energy levels decreases and Bandgap increases. As the Bandgap increases the properties of the material changes.

1D nano material!



Electron can move inside the well but not move outside

Schrodinger equation;

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi(x) = 0$$

Inside the 1-D well: $[V=0]$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} (E) \psi(x) = 0$$

$\rightarrow k^2$

$$k^2 = \frac{2m}{\hbar^2} E$$

$$k = \frac{\sqrt{2mE}}{\hbar}$$

$$E = \frac{(k\hbar)^2}{2m}$$

$$\frac{d^2\psi}{dx^2} + k^2 \psi(x) = 0 \rightarrow \text{solution}$$

$$\psi(x) = A \sin kx + B \cos kx$$

If $x=0 \rightarrow \psi(x)=0$

$$\psi(0) = A \sin k(0) + B \cos k(0)$$

$$0 = B$$

$$\psi(x) = A \sin kx$$

If $x=a \rightarrow \psi(a)=0$

$$0 = A \sin k(a) + 0$$

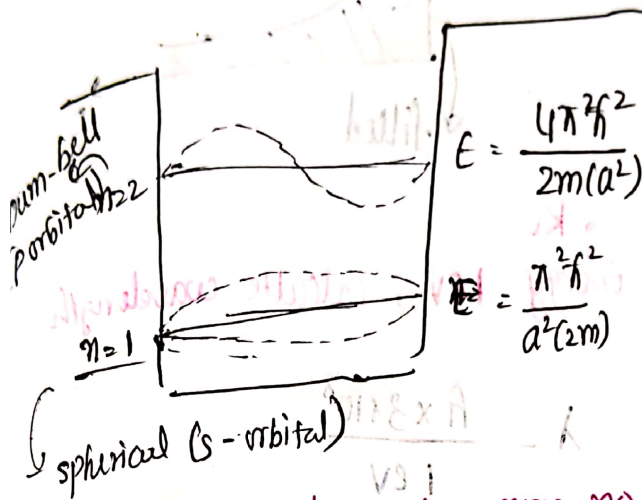
$$\sin k(a) = 0 \rightarrow A \neq 0$$

$$ka = n\pi$$

$$k = \frac{n\pi}{a}$$

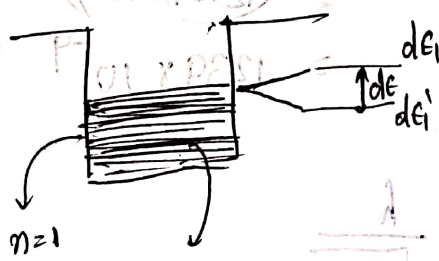
then energy $E = \frac{(\pi\hbar/a)^2 \hbar^2}{2m}$

$$E = \frac{n^2 \pi^2 \hbar^2}{2m a^2}$$



As we are studying for more no. of atoms we will have more no. of states at "equidistance" of the energy level.

$dn \Rightarrow$ no. of allowed energy states given the study/reference



Density of the states

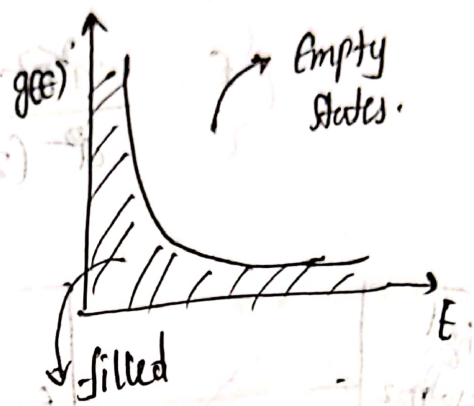
$$g(E) = \frac{dn}{dE}$$

$$dE = \frac{2n dn \pi^2 \hbar^2}{2m a^2} \rightarrow \frac{dE}{dn} = \frac{n \pi^2 \hbar^2}{m a^2}$$

$$g(E) = \frac{dn}{dE} = \frac{m a^2}{n \pi^2 \hbar^2} \rightarrow \frac{m a^2}{\pi^2 \hbar^2} \frac{\pi \hbar}{a \sqrt{2mE}}$$

$$g(E) = \frac{dn}{dE} = \frac{a}{\pi \hbar} \sqrt{\frac{m}{2E}}$$

$$g(E) \propto E^{-1/2}$$



Q: An electron with ^{KE} Energy = 1 eV, calculate wavelength

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{h \times 3 \times 10^8}{1 \text{ eV}}$$

$$\lambda = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{1.602 \times 10^{-19}} \Rightarrow \frac{6.62 \times 3 \times 10^{-7}}{1.602}$$

$$\lambda = 1240 \text{ nm}$$

Calculate $a = ?$

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

MANISH:-

- C
- Si
 - 2nd largest available element
 - Temperature stability.
 - oxidizes easily (SiO_2) and SiO_2 is stable also
 - Environmental safe
 - Si in elemental semiconductor is stable
- Ge
- Sn
- Pb

1st - Oxygen

3rd - Iron

Si $\rightarrow$ 150°C - 200°C

Ge $\rightarrow$ 75°C - 160°C

Direct bandgap & Indirect bandgap semiconductors?

(Si) $\rightarrow$ Indirect bandgap (emits heat)

GaAs $\rightarrow$ High speed (Direct bandgap) (emits IR light)

SiC, GaN

LED

MEMS $\rightarrow$ micro electronic mechanical systems.

mobility $\rightarrow$ Si (μ_n) = 1500 $\text{cm}^2/\text{V-s}$

Ge (μ_n) = 4000 $\text{cm}^2/\text{V-s}$

GaAs (μ_n) = 8600 $\text{cm}^2/\text{V-s}$

$$\mu = \frac{dv}{dE} \rightarrow \text{Electric field}$$

drift velocity

$$v = \frac{q E_s \tau}{m^*}$$

NISHAD:-

Energy levels & Quantum Confinement effects:-

KE	λ	a
electron 1 eV	1.22 nm	0.613 nm
7 eV	0.55 nm	0.274 nm

Quantum tunnelling effect As we confined the electron in the potential well but its wave can travel out of the "potential well".



Interference of waves

$\lambda \sim a$ → De-Broglie's wave effect
 → Then it is called as quantum confinement effect.

Density of states?

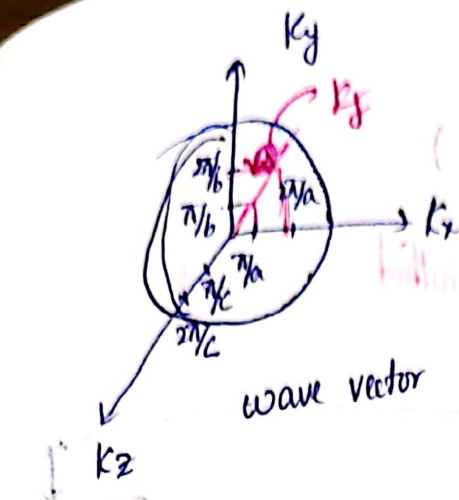
$$g(E) = \frac{dn}{dE} = \frac{m a^2}{\pi^2 \hbar^2} \frac{\pi \hbar}{a \sqrt{2mE}} = \frac{m a}{\pi \hbar \sqrt{2mE}}$$

$$= \frac{1(m) (0.613 \times 10^{-9})}{\pi \hbar \sqrt{2(m)(1eV)}}$$

Density of states for 3D:

WKT: $k = \frac{n\pi}{a}$; $E = \frac{\hbar^2 k^2}{2m} \Rightarrow k = \frac{\sqrt{2mE}}{\hbar}$

Wave vector.



no. of states

no. of states =

$$\frac{4/3 \pi (k_F)^3}{(2\pi/a)^3}$$

$$(a=b=c)$$

Consider any unit cell.

no. of states

$$= \frac{\text{Vo. of sphere}}{\text{Vo. of single cell}}$$

$$= \frac{1}{6} \frac{k_F^3 a^3}{\pi^2}$$

$$\Rightarrow \frac{1}{6} \left(\frac{\sqrt{2mE}}{\hbar} \right)^3 \frac{a^3}{\pi^2}$$

$$\frac{a^3 (2mE)^{3/2}}{6 \hbar^3 \pi^2}$$

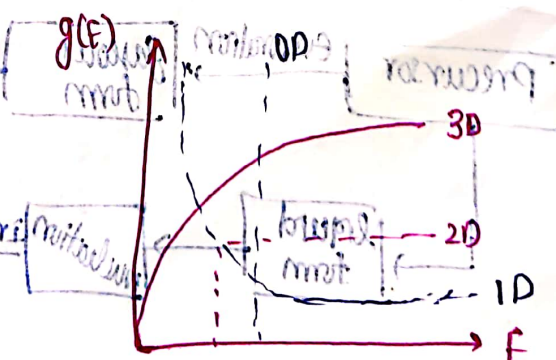
$$\frac{dn}{dE} = \frac{a^3}{2 \hbar^3 \pi^2} \left(\frac{1}{2} \right) E^{1/2} (2m)^{3/2} \rightarrow \text{multiply by 2}$$

$$\frac{dn}{dE} = \frac{1}{2} \frac{a^3}{\hbar^3 \pi^2} (2m)^{3/2} E^{1/2}$$

$$g(E) = \frac{dn}{dE} = \frac{a^3 (2m)^{3/2} E^{1/2}}{(2) \hbar^3 \pi^2} \rightarrow \text{Density of states}$$

3D case

$$g(E) \propto E^{1/2}$$

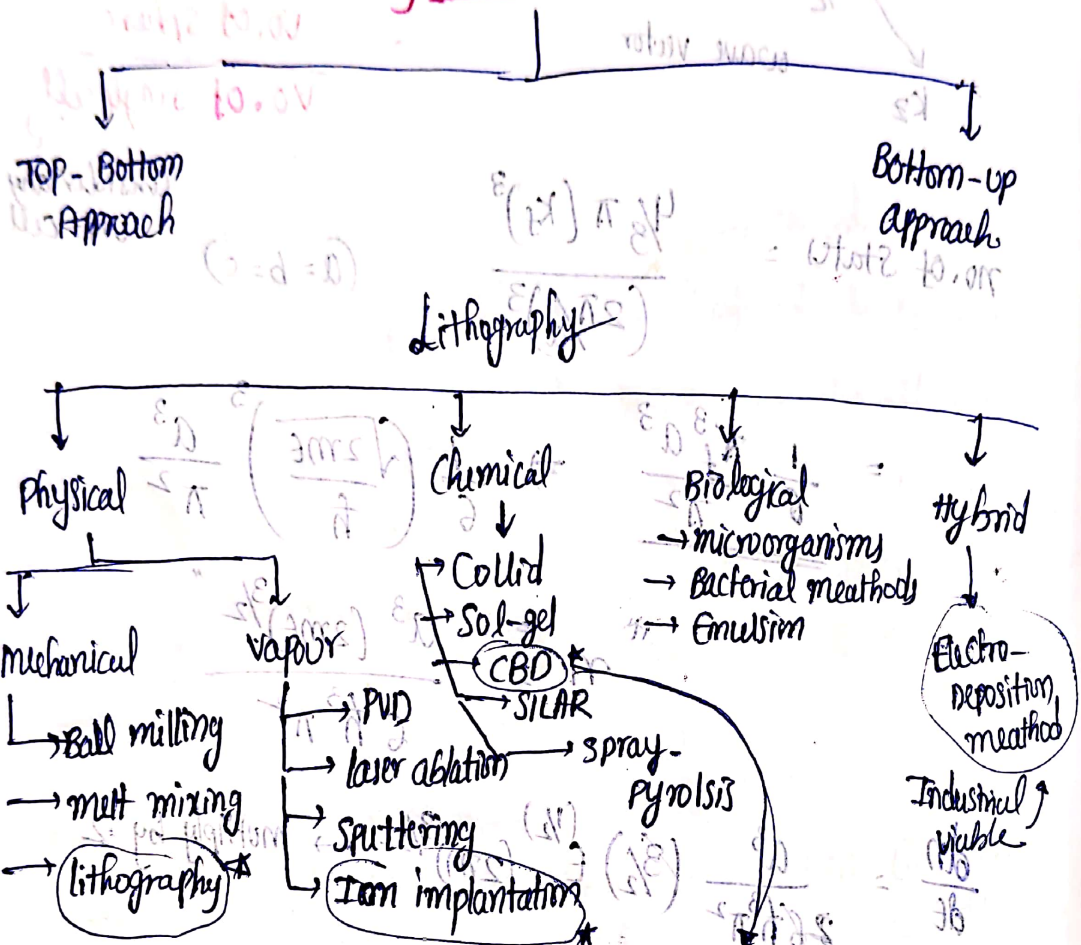


Home work
Do it for 2D, 0D

for 2D $\rightarrow g(E) \propto E^0$

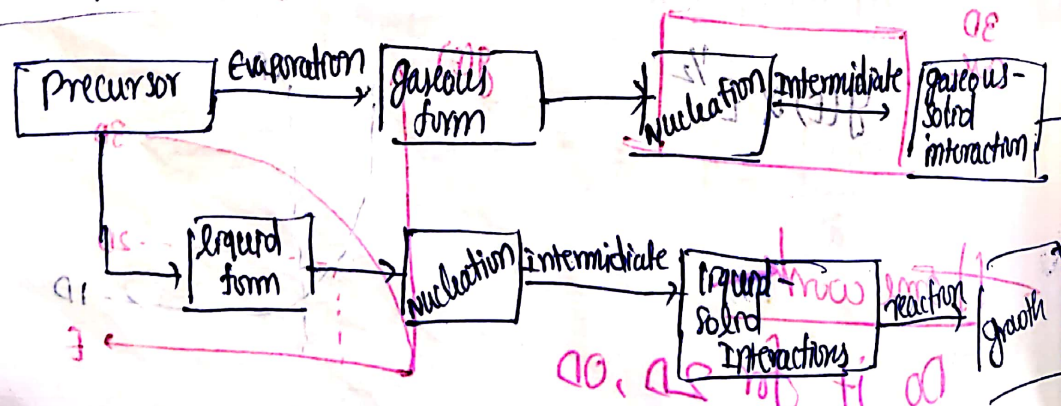
for 3D $\rightarrow g(E) \propto 2\delta(E-E_c)$

Synthesis method

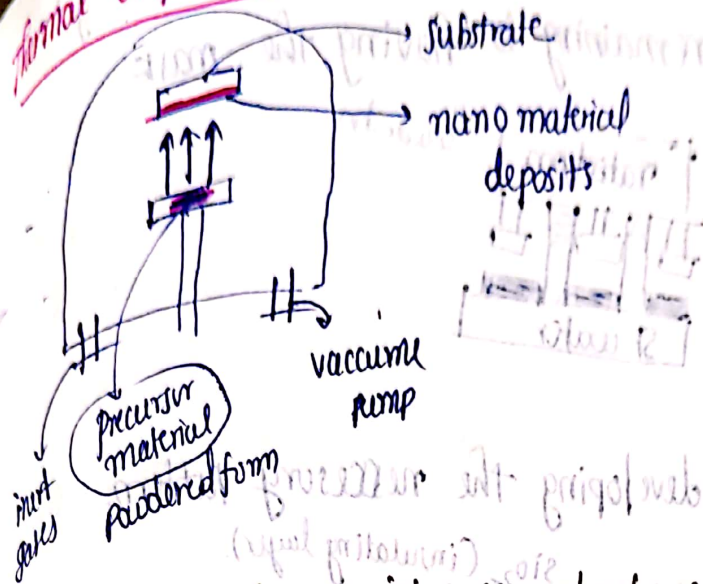


Principles for synthesis method:

- $\rightarrow$ Precursor, evaporation/liquid/gas.
- $\rightarrow$ Nucleation.
- $\rightarrow$ Growth

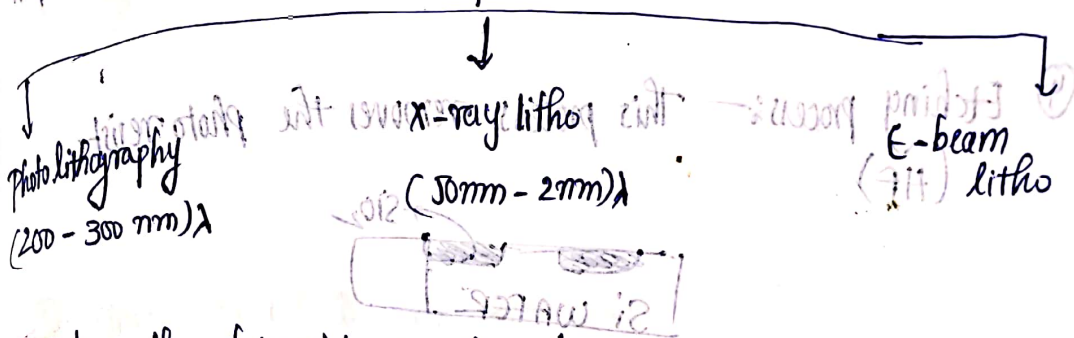


Thermal evaporation method :

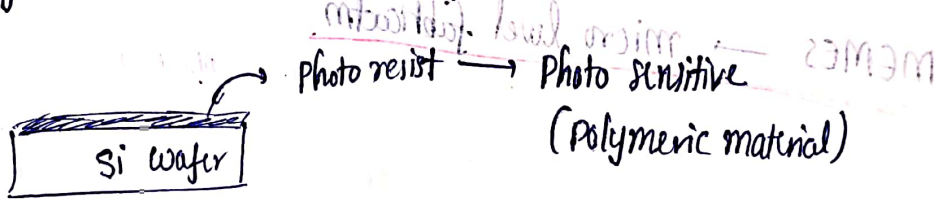


Lithography process : devising nano structures —

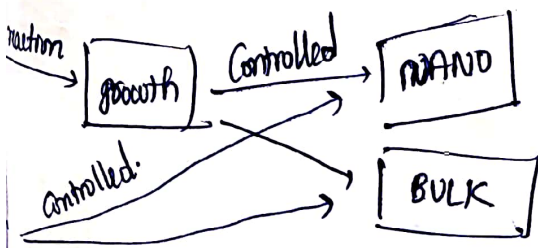
↳ transferring of geometric patterns to develop nano structures.



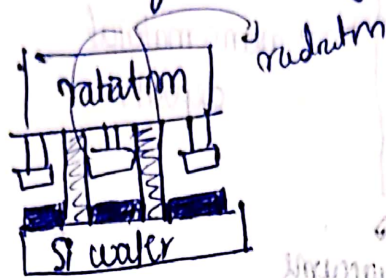
① Coating the photoresist on Si wafer.



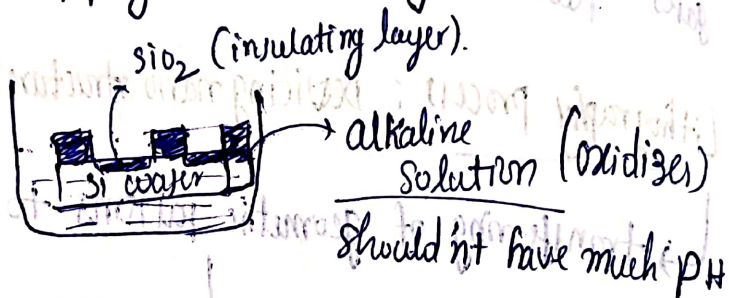
PTD →



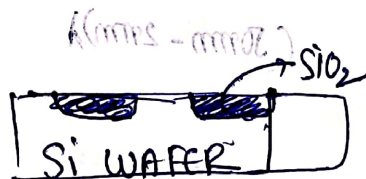
② Exposure to radiation: So that the interested region gets exposed and the remaining is having the mask.



③ Use developer for developing the necessary pattern.

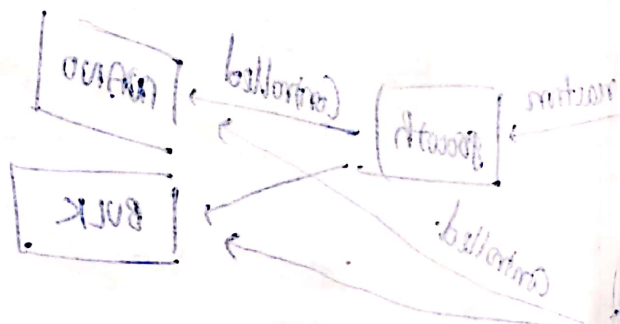
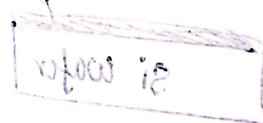


④ Etching process:— This process removes the photo resist (HF)



MEMES → micro level fabricatm

(long term disassembly)



MANISH

Crystal growth & wafer preparation

→ Polycrystalline Silicon (Si) is used as the starting material for the growth of device quality single crystal Si.

Silicon is of 3-types:

① Amorphous Si: NO crystal order, more defects.



② Polycrystalline Si, Multiple crystal order



③ Single crystal Si: Uniform crystal order



Quartzite (Purity of form of Sand SiO_2)

↓
metallurgical grade Si (mgs)

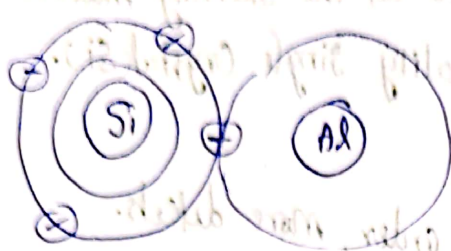
↓
EGS (Electronic grade Si)

↳ It is highest purity of

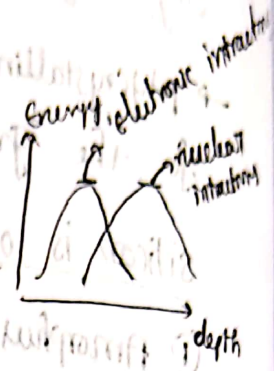
polycrystalline ~ 99.9% pure

NISHAD:

Ion Implantation:



Si Al^+



⊛ When Al^+ moves with very high speed then it is called as Swift heavy ion process.

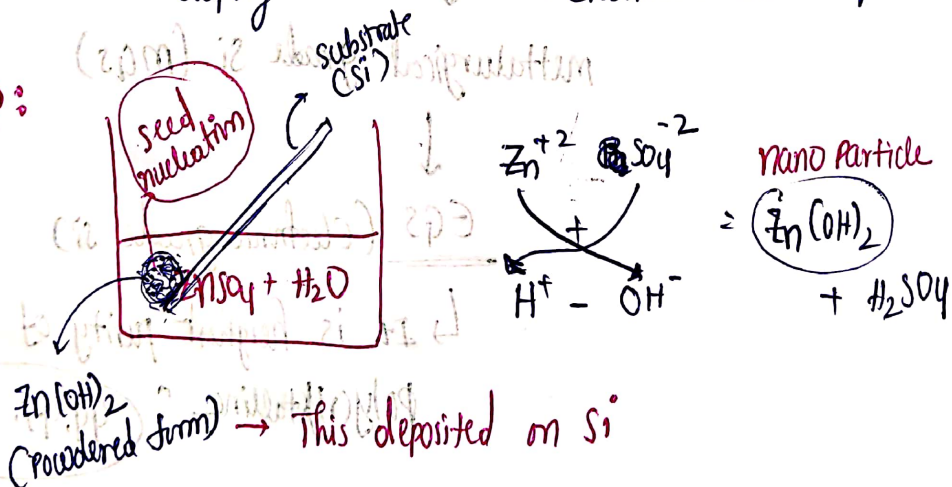
⊛ When we give sufficient energy to Al^+ so that it just stops after interaction with Si then it is called as Ion implantation method.

Al^+ experiences ① Electronic energy loss

② Nuclear energy loss.

(Simple process (But not so accurate) in place of ion implantation method for doping is "CBD" → Chemical Bath Deposition.

CBD:



When we heat the "sol" then $Zn(OH)_2$ goes oxidation & gives "ZnO"

• we can also dope another mono particles by changing the chemical and also in what ratio we are using them for deposition.

BiCMOS: BJT + MOS

proposed TAFIOM of 2014: TAFIOM must meet a TIR ←
proportionate & proportionate at unit level if = groups

Marish:

Single crystal silicon is prepared using following 3 techniques:

① Czochralski method: most used, used for both Si & GaAs.

② —

③ —

④ Asymmetric! write the fabrication steps of N-MOSFET

Bicmos: BJT + MOSFET

Inverter using BJT, CMOS, ~~BiMOS~~ BiCMOS.

→ BJT is faster than MOSFET; Due to MOSFET having Capacitor → it takes time for charging & discharging

NISHAD:
SRM/TRIM $\rightarrow$ Software for stopping potential (ion implantation method).

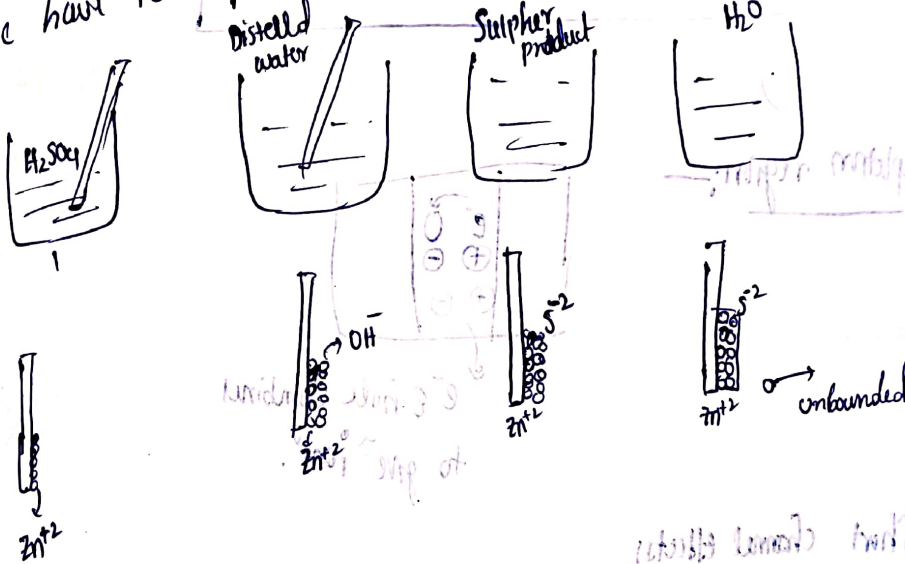
Ostwald's ripening?

- Concentration precursor sol
 - volume of sol
 - temperature
- PH
- time
- } these control the "CBO"

SILAR (Successive ionic layer adsorption and reaction).

we have to dip the rod successivly into the beakers.

distilled water Sulphur product H_2O



Chemical Substances
 based on the pH of the solution
 - If pH is below 7
 - based on pH of the solution
 - If pH is above 7

brunni blauergrün - dol. 1
 1/2 - 1/3 spezif. blauergrün - s.

11/11/1919

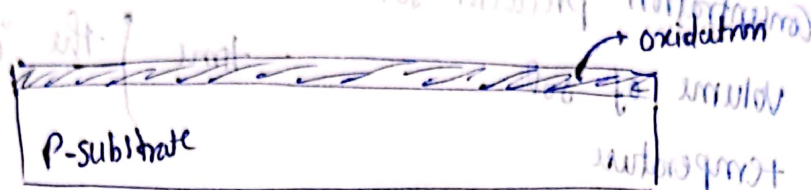
above level (A) standard (1)
above level (A) positive (2)

Bicmos: BJT + mosfet (faster than mosfet).

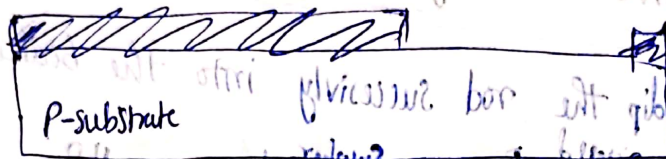
Step-1:



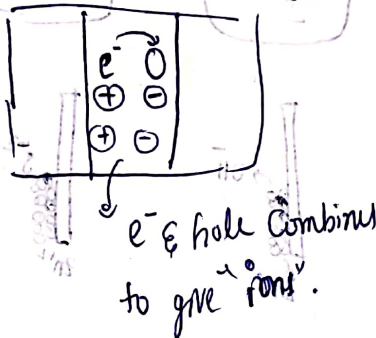
Step-2:



Step-3:



Depletion region:



⊗ Short channel effects

1. Sub-threshold current
2. Threshold voltage roll-off
- 3.
- 4.
- 5.
- 6.
- 7.

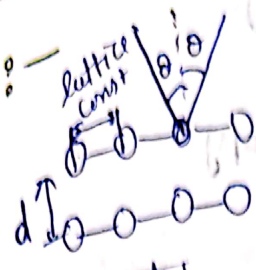
Sub-threshold currents

- ① $V_{GS} < V_{th}$ then current should be off
- ② BUT some diffusion current flows due to short channel effect

VLSI Design layout rules:

- ① lambda (λ) based rules
- ② micron (μ) based rules.

NISHAD:



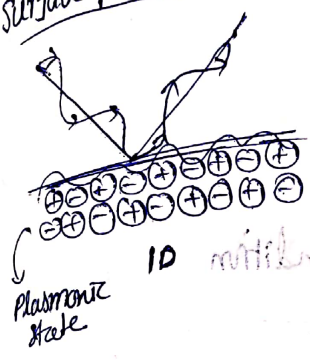
$$2d \sin \theta = n\lambda$$

NANISH:

VLSI layout stick diagram:
wiring (Interconnection):

NISHAD:

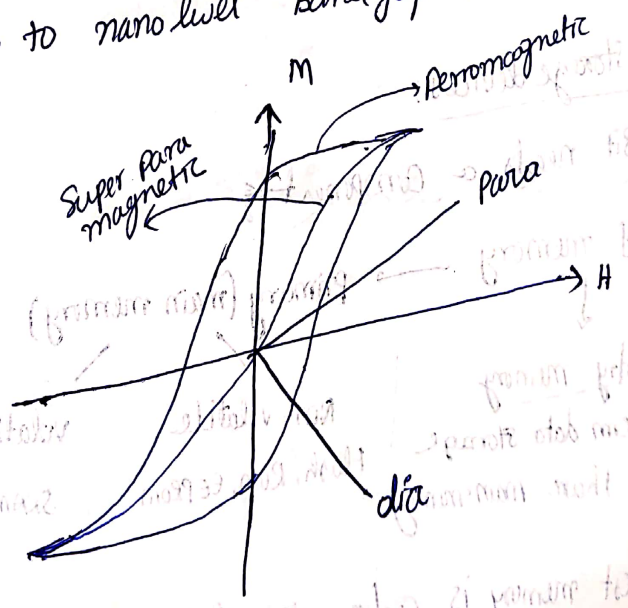
Surface plasmonic effect:



Here if the frequency of light matches the freq of IO material then resonance happens.

that is why gold (yellow) at nano dimension gives red colour.

→ As we go to nano level Band gap increases



- ① XRD : x-ray diffraction (structural)
- ② XPS : x-ray photoemission spectroscopy
(chemical composition).

Photoelectric effect:

Solar cells!

Photonics: Butterfly wing is made up of periodic arrangement of nano structures.

MANISH:

Spice device models

Thermal run out!

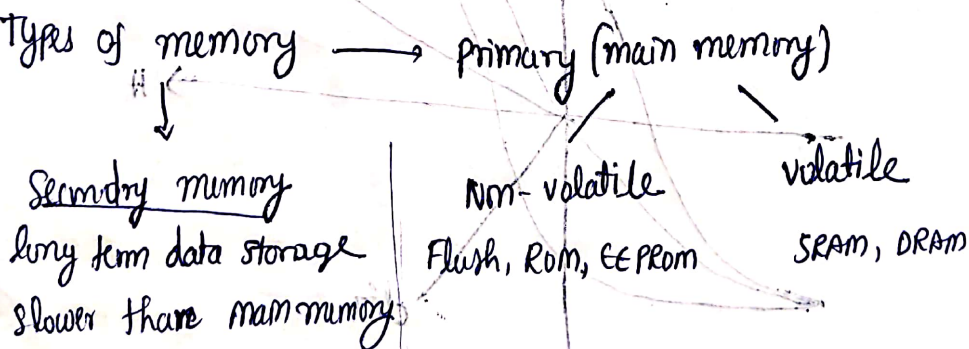
Effect of gate voltage on surface condition

Energy band diagram → study!

Data Storage devices

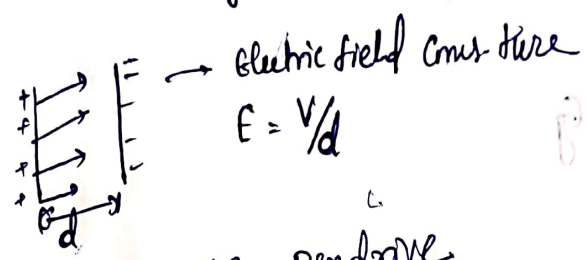
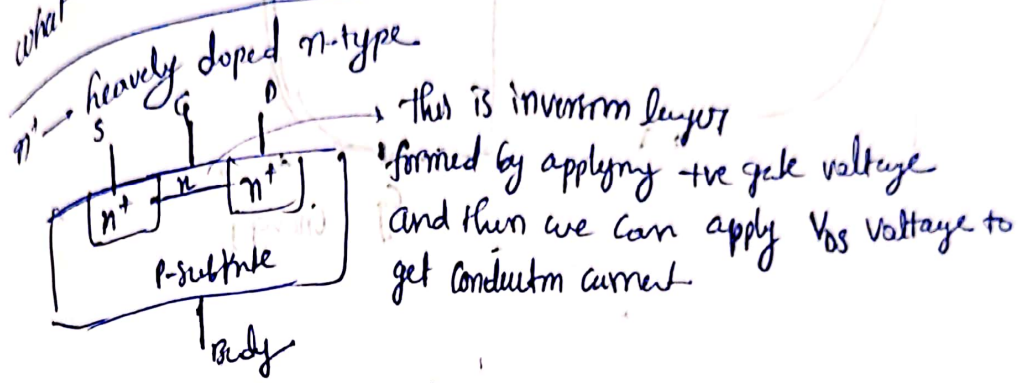
Each Bit needs a Component

Types of memory



→ Fastest memory is cache, after that Rom is fastest

Flash memory (follows the EEPROM mechanism).
 what is this floating gate & control gate?

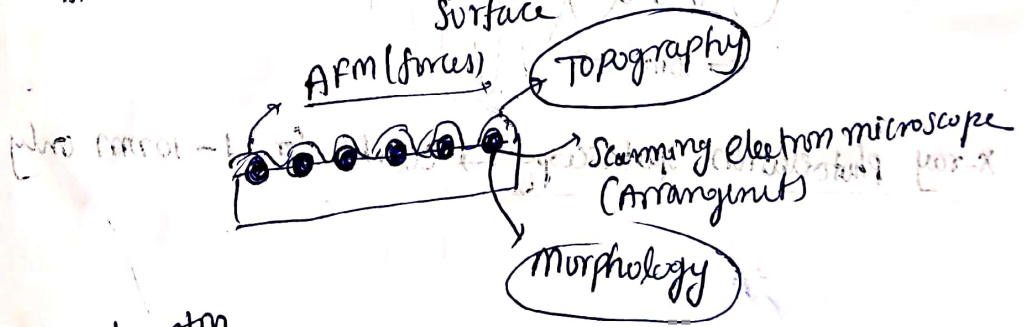


Flash memory: Pen drive

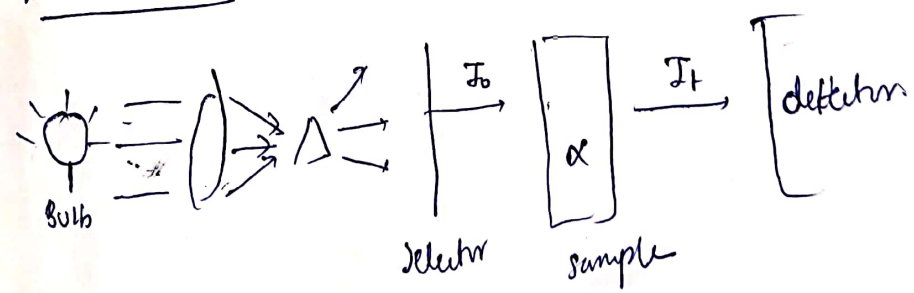
Photocopy

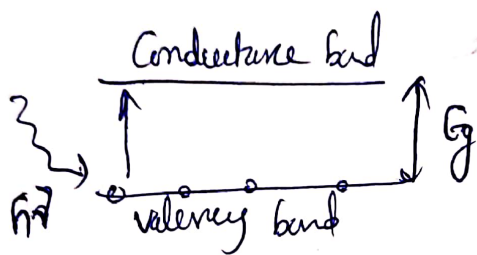
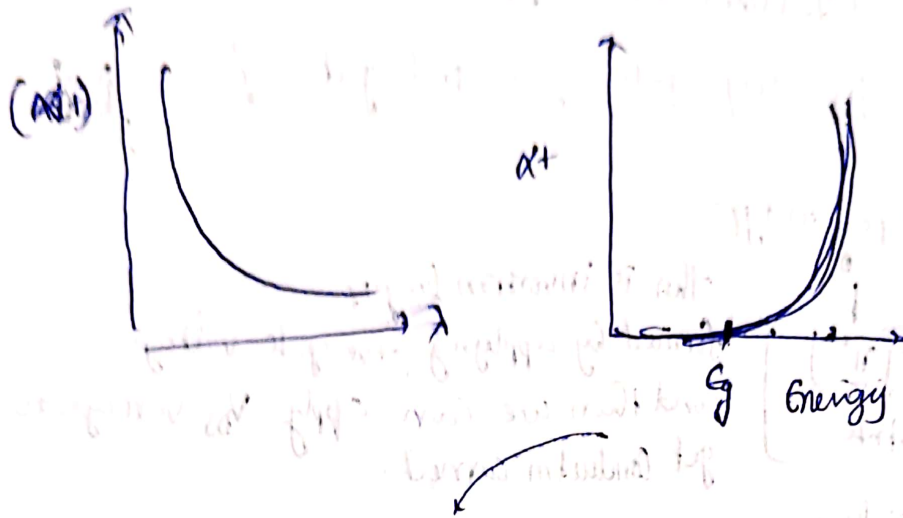
NISHAD

- ① Electron microscopy : we can see the arrangements of atoms/electrons on the surface
- ② Atomic force microscope : we can observe the force on the surface



$\lambda \rightarrow$ absorption

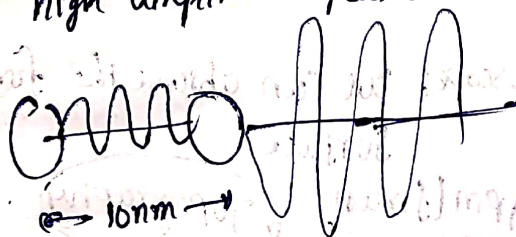




Spectroscopy

Infrared spectroscopy: tells about the bonds & the atoms in the nano particles.

If the IR freq matches the freq of atoms then resonance takes place & gives high amplitude peak



x-ray photoelectron spectroscopy → works for 1-10nm only